INCH-POUND

MS17825H
5 February 2004
SUPERSEDING
MS17825G
30 November 1988

MS SPECIFICATION SHEET

NUT, SELF-LOCKING, CASTELLATED, HEXAGON, .250°F NON-METTALIC INSERT

Reactivate after 5 February 2004 and may be used for new and existing designs and acquisitions.

This specification is approved for use by all Department and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this MS specification sheet and NASM25027.

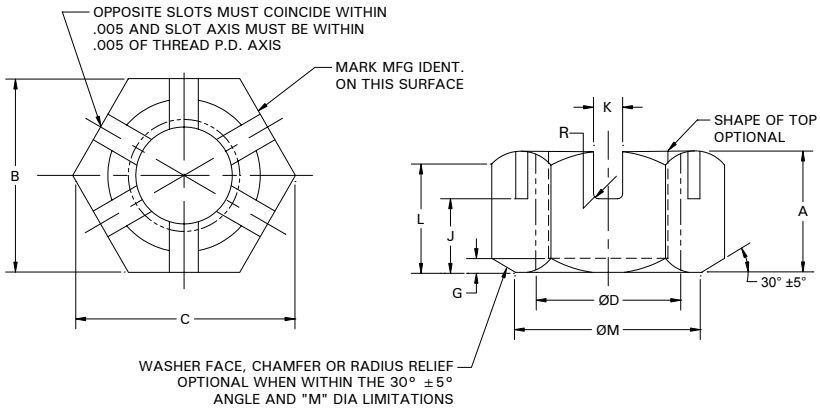


TABLE I. NUT DIMENSIONS AND STRENGTHS

SIZE DASH NO.	THREAD AS8879	A		B		C	ØD	G		J	
		MAX	MIN	MAX	MIN	MIN	MAX	MAX	MIN	MAX	MIN
-3	.1900-32 UNJF-3B	.265	.235	.376	.367	.410	.215	.032	.006	.125	.105
-4	.2500-28 UNJF-3B	.296	.266	.439	.430	.482	.275	.036	.007	.141	.121
-5	.3125-24 UNJF-3B	.343	.313	.502	.492	.552	.338	.042	.008	.188	.168
-6	.3750-24 UNJF-3B	.421	.391	.564	.553	.622	.400			.235	.215
-7	.4375-20 UNJF-3B	.468	.438	.690	.679	.766	.463	.050	.010	.282	.262
-8	.5000-20 UNJF-3B	.578	.548	.752	.741	.837	.525			.375	.355
- 9	.5625-18 UNJF-3B	.624	.594	.877	.865	.978	.587	.055	.012	.406	.386
-10	.6250-18 UNJF-3B	.734	.704	.940	.928	1.051	.650			.484	.464
-12	.7500-16 UNJF-3B	.828	.798	1.064	1.052	1.191	.795	.063	.014	.578	.558
-14	.8750-14 UNJF-3B	.921	.891	1.252	1.239	1.403	.900	.072	.016	.672	.652
-16	1.0000-12 UNJF-3B	1.015	.985	1.440	1.427	1.615	1.025			.766	.746
-18	1.1250-12 UNJF-3B	1.171	1.141	1.627	1.614	1.825	1.150	.083	.018	.828	.808
-20	1.2500-12 UNJF-3B	1.265	1.235	1.815	1.801	2.038	1.275			.891	.871

AMSC N/A

FSC 5310

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TABLE I. (CONT'D)

SIZE DASH NO.	K		L	∅M		R	ULTIMATE TENSILE STRENGTH LB MIN	TEST WRENCHING TORQUE (INCH LBS)	ASSEMBLY TORQUE VIBRATION (IN. LBS)	WEIGHT LB/100
	MAX	MIN	MIN	MAX	MIN	MAX		MIN	MAX	MAX
-3	.108	.078	.110	.376	.355	.035	1,950	35	28	.400
-4			.125	.439	.418		3,500	80	65	.650
-5			.172	.502	.480		5,500	225	180	1.000
-6	.155	.125	.218	.564	.543	.053	9,000	325	260	1.400
-7			.265	.690	.675		12,200	575	460	2.000
-8			.359	.752	.737		16,600	900	720	3.900
-9	.187	.157	.390	.877	.862	.062	21,200	1,100	880	6.900
-10			.468	.940	.925		27,400	1,600	1,300	11.400
-12			.562	1.064	1.049		40,000	2,800	2,200	16.900
-14			.656	1.252	1.237		54,600	4,600	3,700	26.000
-16			.750	1.440	1.425		71,000	6,800	5,400	38.000
-18			.844	1.627	1.612		92,600	10,000	8,000	53.500
-20			.938	1.815	1.800		118,500	14,000	11,000	74.500

REQUIREMENTS:

MATERIALS: Steel, UNS G11370 (AISI 1137) per AMS5024
UNS G11374 (AISI 11L37) per AMS5020
UNS G41300 (AISI 4130) per AMS6348
UNS G87400 (AISI 8740) per AMS6322, 6323, 6325, 6327, and 6358.
INSERT MATERIAL: Nylon.
FINISH: Cadmium plating in accordance with SAE-AMS-QQ-P-416, Type II, Class 2.
Break all sharp edges and remove burrs.
SURFACE TEXTURE: 125 in accordance with ASME B46.1.
DIMENSIONS: All dimensions are in inches unless otherwise specified.
DESIGN INFORMATION: These nuts may be used at temperatures 250° F and below in those types of applications for which slotted or castellated nuts are approved for use, and if such applications conflict with any limitation on the use of self-locking nuts, as cited by NASM33588, this standard takes precedence. Nuts are to be installed in accordance with NASM33540.
EXAMPLE OF PART NUMBER: MS17825-4 = .2500-28 nut, cadmium plated, non-metallic insert.
PERFORMANCE: Per NASM25027 except for:
1 – Tabulated ultimate tensile strength values.
2 – Vibration per NASM1312-7, (30,000 cycles) using assembly torques listed in Table I for one seated cycle.
3 – Locking torque valves applicable for one cycle at room temperature only.
Certain provisions (limits for across flat dimension) of this standard are the subject of International Standardization ABC STD 17/2. When revision or cancellation of this standard is proposed which will affect or violate the international agreement concerned. The preparing activity will take appropriate reconciliation action through international standardization channels, including departmental standardization officer, if required.
For design feature purposes, this standard takes precedence over procurement documents referenced herein. Referenced documents shall be of the issue in effect on date of invitations for bids, or request for proposal.
Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Navy – AS
Air Force – 99

Preparing Activity
DLA-IS

(Project 5310- 2710)

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